

Distributed Passive SONAR Systems for Target Localization

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Abstract

The development of underwater surveillance systems has become increasingly important due to recent events. One approach is a passive SONAR system, which enables the detection of targets without being noticed. To improve the performance of target detection, distributed sensor systems can be used. By adding further subsystems (nodes), it is possible to achieve more precise detection. In this context, so-called *ad-hoc systems* with an arbitrary distribution of passive, static underwater sensor nodes are investigated. Precise localization is made possible by knowing the position and orientation of the individual sensor nodes and by suitable signal processing. In addition, all sensor systems include activity detection to detect possible targets. This article presents the algorithms developed for real-time target detection and localization. Additionally, the algorithms are tested in a simulation environment and the results are analyzed and discussed.

Introduction

The attacks on the Nord Stream 2 and earlier reports of the destruction of sea cables in the Baltic Sea [1] show the importance of underwater surveillance. To ensure harbor and ocean surveillance, it is required to design suitable systems to detect and localize unauthorized targets. One method of underwater detection is the use of passive SONAR, which gathers all possible target information by analyzing received underwater sound transmissions. To enhance information acquisition, distributed systems will be used. By detecting potential targets and estimating the Direction of Arrival (DOA) simultaneously with more than one system, the position of the targets can be determined. For real-time monitoring, the systems are implemented as real-time applications within the *Kiel Real-time Application Toolkit* (KiRAT) framework. It has an extensive digital oceans simulation to test and evaluate the algorithms that are later used in real systems. The paper is divided into a description of the distributed passive SONAR system, followed by a simulation, and an evaluation using the digital ocean.

Distributed Systems

Distributed systems are a group of independent systems that fulfill a defined task. There are different topologies of distributed systems, which have various advantages and disadvantages in terms of complexity, number of nodes, and the task to be accomplished [3]. The topology chosen for the target localization is called a *peer-to-peer* architecture, which is a decentralized topology.

Each node processes information and sends it to every other node in the network, enabling direct communication as an advantage. Additionally, the network is more stable against the failure of a single system, because the other nodes can still accomplish their tasks. A disadvantage is the complexity of the network, which increases as more nodes are added.

An overview of the node structure of the distributed system with its main functions is in Figure 1.

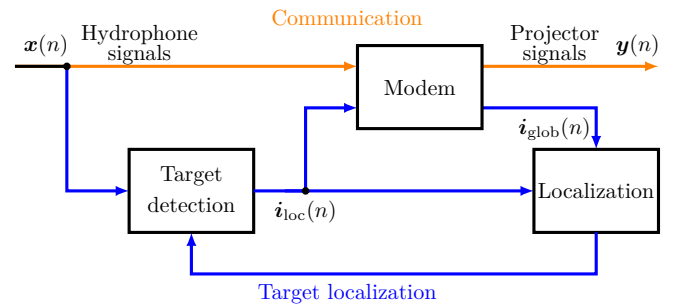


Figure 1: Structure of a node in the distributed system divided into communication and target localization. Target detection is processed independently, while localization is performed using information from other nodes as well.

As a part of underwater target detection, the nodes are placed underwater and get hydrophone signals $x(n)$ as input. The system is structured in two main tasks: communication and target localization, with the latter being the primary objective. In terms of communication, the nodes require underwater communication to exchange information, which is done through the underwater channel. A modem filters information signals by hydrophone input and sends generated information via projectors to the underwater channel. However, since the focus is on target localization, communication is currently assumed to be ideal, in the sense of assuming no errors. To accomplish target localization, two key steps of target detection and position estimation are required and are described in the following sections.

Target Detection

In the target detection process of each node, a large amount of information is gathered as part of the passive system. Therefore, only received data can be utilized. Figure 2 shows the structure of the target detection process along with its main tasks. To define a detected target, two steps of activity detection and DOA estimation must be performed.

Both tasks are designed to obtain the short-term spectrum of the hydrophone signals as inputs. Therefore, the hydrophone signals will be transformed using the analysis filter bank, to obtain the spectrum of each hydrophone signal $\mathbf{X}(e^{j\Omega_\mu}, n)$ at specific time points n . As a first step, only a single target detection is analyzed.

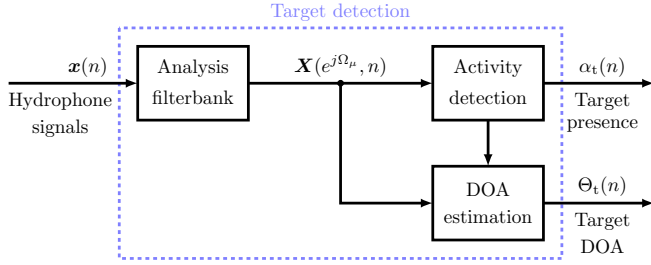


Figure 2: Target detection, divided into activity detection and DOA estimation, with input from the hydrophones frequency responses of the analysis filter bank.

Activity Detection

The goal of activity detection is to identify a possible target and confirm its presence. A possible target is analyzed based on the received acoustic signal level. If a target moves closer to a node, the received signal strength increases over time. To track this, the activity detection stores time-averaged signal strengths. If these values increase monotonically, a possible target is identified by the node. With

$$\beta_{Pt}(n) = \begin{cases} 1, & \text{if } f(\bar{X}(e^{j\Omega_\mu}, n)) \geq f(\bar{X}(e^{j\Omega_\mu}, n-1)). \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

a possible target is detected using linear regression $f(\dots)$ over the average of all hydrophone signal strengths $\bar{X}(e^{j\Omega_\mu}, n)$. Once a possible target is identified, the DOA estimation starts to analyze.

DOA Estimation

To estimate the DOA of a target using only received signals, various algorithms can be applied. In real-time systems, it is important to use suitable algorithms. A simple and efficient approach is to perform beamforming across all hydrophones of the node. Using the *delay-and-sum* method, a beamformer can be computed by weighting each hydrophone signal with a steering vector $\mathbf{a}_\mu(\phi)$ at a given frequency point μ . Since the target's frequency response is unknown, multiple frequency points are analyzed. The beamformer is determined using the norm power of N hydrophone signals and M frequency points:

$$b(\phi, n) = \left| \frac{1}{M} \frac{1}{N} \sum_{\mu=0}^{M-1} \sum_{i=0}^{N-1} a_{\mu,i}(\phi) \cdot X_i(e^{j\Omega_\mu}, n) \right|^2. \quad (2)$$

The DOA of the target is then determined with

$$\Theta_t(n) = \underset{\phi}{\operatorname{argmax}} \{b(\phi, n)\}, \quad (3)$$

since the highest signal strength correspond to the transmission angle of the target.

Localization

After calculating the DOA of each node and exchanging information among them, the target's location can be determined. The position and orientation of the nodes are known at the beginning of each measurement. Using this information and also the DOA estimation of the target, localization is performed via triangulation. For each node a straight line can be expressed as

$$\mathbf{k} = \underbrace{\begin{bmatrix} x_s \\ y_s \end{bmatrix}}_{\mathbf{k}_0} + \alpha \underbrace{\begin{bmatrix} \frac{\cos(\Theta_t(n))}{|\cos(\Theta_t(n))|} \\ \frac{\sin(\Theta_t(n))}{|\sin(\Theta_t(n))|} \cdot \tan(\hat{\Theta}_t(n)) \end{bmatrix}}_{\mathbf{k}_1}, \quad (4)$$

where the position vector $\mathbf{k}_0$ is defined by the 2D coordinates x_s and y_s of the node. The direction vector is calculated using the relative incidence angle $\hat{\Theta}_t(n)$ and the sign of the DOA angle in the x- and y-coordinates. After the straight lines are calculated, the target's position is estimated by finding their intersection points. This is done by expressing two straight lines $\mathbf{k}$ and $\mathbf{l}$ as a linear equation system. To solve this linear equation system numerically, Gaussian elimination is applied. For that, the linear equation system will be rearranged like

$$\underbrace{[\beta l_1 \quad -\alpha k_1]}_C = \underbrace{[\mathbf{k}_0 - \mathbf{l}_0]}_d, \quad (5)$$

to express it in matrix form. The algorithm determines the intersection point using the variables $\hat{\alpha}_t$ and $\hat{\beta}_t$ for each straight line. Inserting one determined variable to the corresponding straight line equation system leads to an estimate of the target position:

$$\hat{\mathbf{k}}_t = \mathbf{k}_0 + \hat{\alpha}_t \cdot \mathbf{k}_1. \quad (6)$$

In a distributed system with more than two sensors, multiple intersection points $\mathbf{c}_{i,j}$ are calculated. Even with small errors in DOA estimation, differences in the estimated target's position occur for each intersection point. To improve accuracy, the target position is estimated using all L possible intersection points with an additional weighting factor w_i . A method of weighting is based on the signal strength of the nodes. Higher signal strength leads to a DOA estimation's with a higher resolution, so intersection points from these are more weighted. With

$$\mathbf{p}_t = \frac{\sum_{i=0}^{L-2} \sum_{j=i+1}^{L-1} \mathbf{c}_{i,j} w_i w_j}{\sum_{i=0}^{L-2} \sum_{j=i+1}^{L-1} w_i w_j} \quad (7)$$

the final target's position is estimated with each weighted intersection point.

Additionally, to reduce computational effort during processing, sub-angle analysis can be utilized. In DOA estimation, it's unnecessary to calculate the entire beam pattern due to the estimation is based on the highest value. Therefore, restricting the estimation of a specific area of angles can significantly decrease the number of calculations. To define the specific area and enable sub-angle analysis involving two steps. First, the DOA of a

possible target needs to be calculated to set the center of the sub-angle area. Then, by storing more than two position estimations, an estimated velocity of the moving target $\hat{v}_{t,n}$ can be calculated. The area of the next possible DOA can then be limited with the next maximum DOA by

$$\Theta_{\max} = \arcsin\left(\frac{\hat{v}_{t,n}\delta}{\hat{d}_{t,n}}\right), \quad (8)$$

with $\hat{d}_{t,n}$ represents the distance between a node and the target. The product of the estimated velocity $\hat{v}_{t,n}$ and the time interval between two measurements δ , defines the highest possible distance that the target can travel and is used for the limitation. Consequently, the number of analyzed angles is restricted twice the maximum calculated DOA $\Theta_{\max}$. The sub-angle estimation is updated after every estimation cycle. To avoid large outliers, a smoothing method is applied to the previously calculated maximum DOAs. If no suitable DOA is obtained through sub-angle analysis, the entire beamformer is computed until the next reliable DOA is identified.

Simulation

As previously mentioned, the distributed system with its functionality, shown in Figure 1, will be implemented as a real-time system. To simulate and integrate the entire system, it is developed within a real-time signal processing framework. It includes a comprehensive digital ocean that simulates underwater sound propagation. Additionally, various types of noise, such as wind strength, additive white Gaussian noise and more can be introduced into the simulation. Furthermore, it is possible to construct arbitrary sensor systems, which can be randomly placed within the digital ocean. Targets can be simulated using sound transmission based on recorded underwater ship noises and predefined trajectories. Figure 3 shows the visualization of the digital ocean, where a target is represented by a submarine and three sensor systems receiving hydroacoustic signals as input.

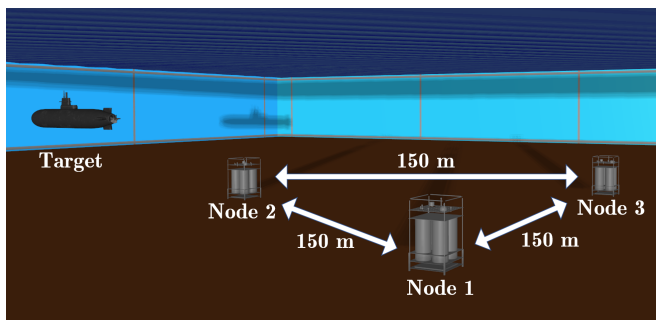


Figure 3: An excerpt from the digital ocean in KiRAT. The target is represented by a submarine and the three nodes as sensor systems.

Simulation Parameter

The sensor systems are simulated as nodes positioned on the ocean floor. Each node has five hydrophones and their geometrical arrangement is shown in Figure 4. This specific arrangement of hydrophones is chosen because sensor systems with this architecture have already been realized and used for real measurements. In [2], several

measurements were tested and evaluated using a passive DOA algorithm. Therefore, testing the system in simulation ensures that it can be directly applied to real-world measurements and compared accordingly.

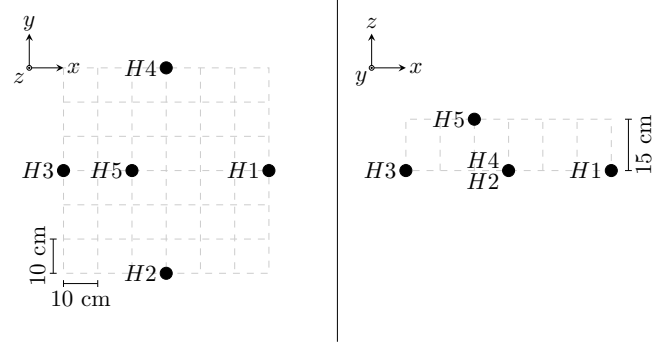


Figure 4: Geometrical arrangement of hydrophones in the node of the distributed system, adapted from [2].

Regarding the distributed system, it has already been mentioned that the communication is assumed to be ideal. Consequently, there is no time delay in the transmission or reception of information between nodes. For tracking purpose, a time interval of 5 s is used to store the position estimations of the nodes.

The number of nodes is currently set to three, but could be easily increased. To achieve the best possible target tracking results, the nodes are placed in an equilateral triangle. This configuration ensures equal spacing between each node, optimizing the calculation of all intersection points from all nodes.

Evaluation

To demonstrate the functionality of activity detection, DOA estimation and target localization under various conditions, the target's trajectory includes linear and round paths. Furthermore, the target moves towards and away from the nodes to simulate variations in the received signal strengths.

Figure 5 illustrates tracking results of a moving target exemplary with constant velocity of $5 \frac{m}{s}$. The direction of the target is represented by an arrow above the trajectory. Estimated positions are marked in red, while the corresponding real positions are depicted as circles. At the beginning of the tracking process, the target remains undetected due to its high distance to the nodes. As the target approaches, the received signal strength increases, allowing the distributed system to identify a possible target and obtain reliable DOA estimates.

During the linear path of the trajectory, the tracking closely follows the target's movement, though positioning errors occur at measured time points. The accuracy improves as the target gets closer to the nodes, which is obtained in the round path of the trajectory. However, after the target completes the turn and begins moving away, the tracking gets worse. Positioning errors increase, although the movement pattern remains recognizable. As the target moves away from the sensor system, the re-

ceived signal strength weakens, preventing reliable DOA estimations. Consequently, at the end of the simulation, the target is no longer detectable due to a high distance. Additionally, significant position estimation errors occur when the target's position aligns linearly with a sensor pair. The more collinear the target is with two sensors, the lower the tracking accuracy.

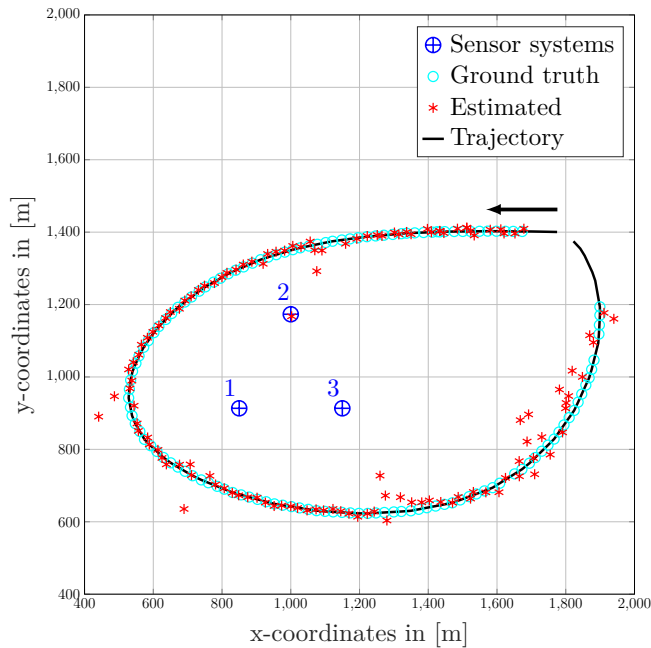


Figure 5: Result of target localization (red) by three nodes. The target's trajectory and direction are shown in black.

Figure 6 compares the DOA estimation of each sensor system to the ground truth from the simulation in Figure 5.

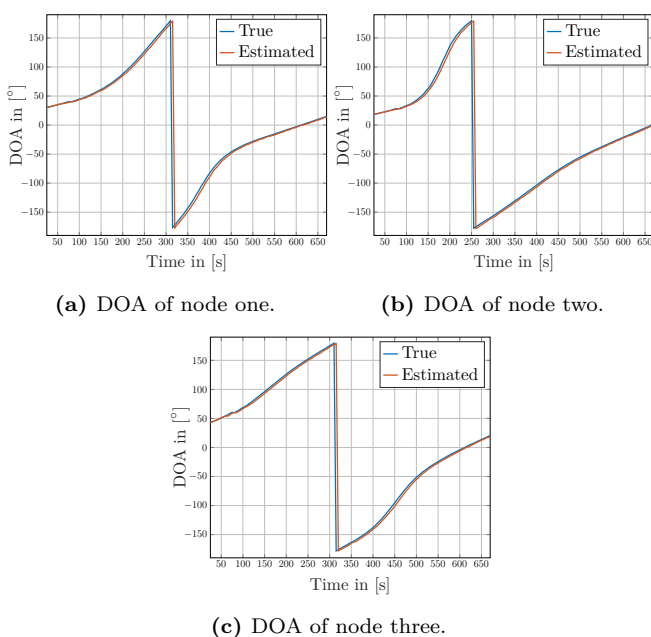


Figure 6: DOA estimations (red) of Figure 5 compared to ground truth (blue).

The root mean square error (RMSE) of the DOA estimations across all nodes is 5.5° , which is not directly visible in Figure 6 due to the small size. The comparison of the DOA estimation to the ground truth shows that the errors remain relatively consistent over time. However, the accuracy of position localizations get worse by increased distance, because the estimation via intersection points. This leads to the conclusion that even small errors in DOA estimation at each sensor node can significantly influence position estimation, as they result in biased intersection points. The overall RMSE of the position estimation is 38.7 m.

Conclusion and Outlook

Overall, a distributed system for passive target detection has been developed. Each node is identical, processing, receiving and transmitting information to fulfill the task of tracking. The structure of the nodes is divided into two main functions: communication and target localization. As a first approach, communication is set to ideal with instant information exchange. Target localization is divided into detection of a possible target and the estimation of its position.

The entire system is implemented as a real-time system and tested in a digital ocean. The results demonstrate that activity detection is possible, and the target is detected only within a specific area. Additionally, estimation accuracy improves when the target is closer to the nodes. Overall, the system successfully enables passive target detection in a distributed setup, providing a solid foundation for further improvements.

As the next step, passive target detection will be extended with a more reliable weighing method. By considering the arrangement of the nodes positions, different intersection points can be weighted accordingly. Additionally, to achieve a more precise determination of the target's position, a suitable Kalman Filter will be implemented for a real-time processing. To further increase realism, a simple underwater communication model will be implemented, allowing the estimated DOA to be transmitted hydroacoustically. Finally, the developed system will be tested in the Baltic Sea to evaluate its performance under real conditions.

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